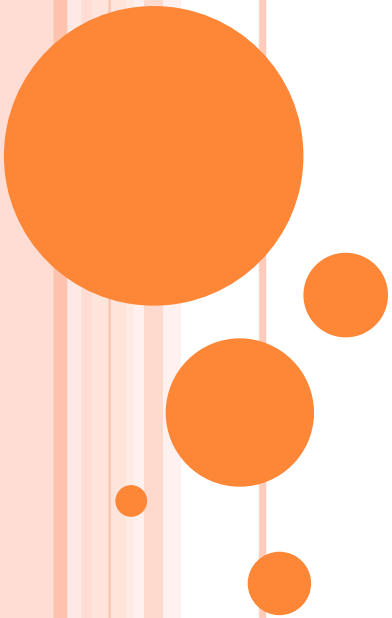
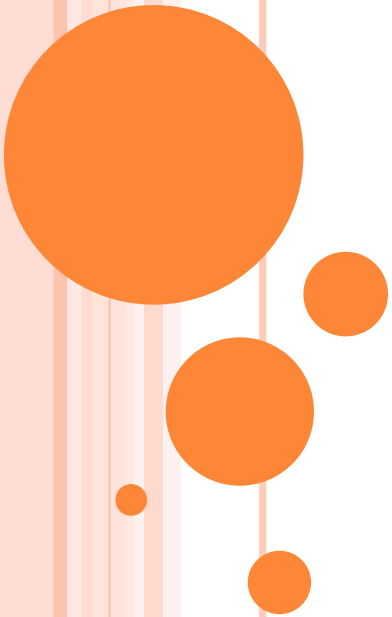
Reduction of Entity-Relationship Diagram (ERD) to Relational Schema



SESSION -1



Reduction of ERD to Schema

- E-R Model and Relational Database Model are abstract, logical representations of real-world enterprises.

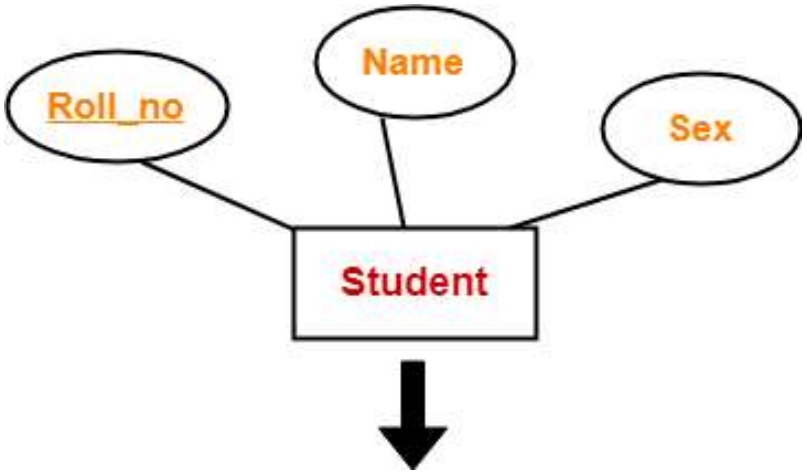
- Because both employ similar design principles, we can convert an E-R design into a Relational Design.



For Strong Entity Set With Only Simple Attributes

- Entity sets which have a primary key are known as strong entity sets.
- A strong entity set with only simple attributes will require only one table in relational model.
  - Attributes of the table will be the attributes of the entity set
  - The primary key of the table will be the key attribute of the entity set





Roll_no	Name	Sex

- Is it a Strong Entity Set?
  Yes
- Contains Simple Attributes?
  Yes
- Any Complex Attributes (Composite, Multi-valued)?
  No

Roll_no	Name	Sex

- Any Key Attribute?
  Always Yes for Strong Entity Set

Schema :

Student (Roll_no , Name , Sex)

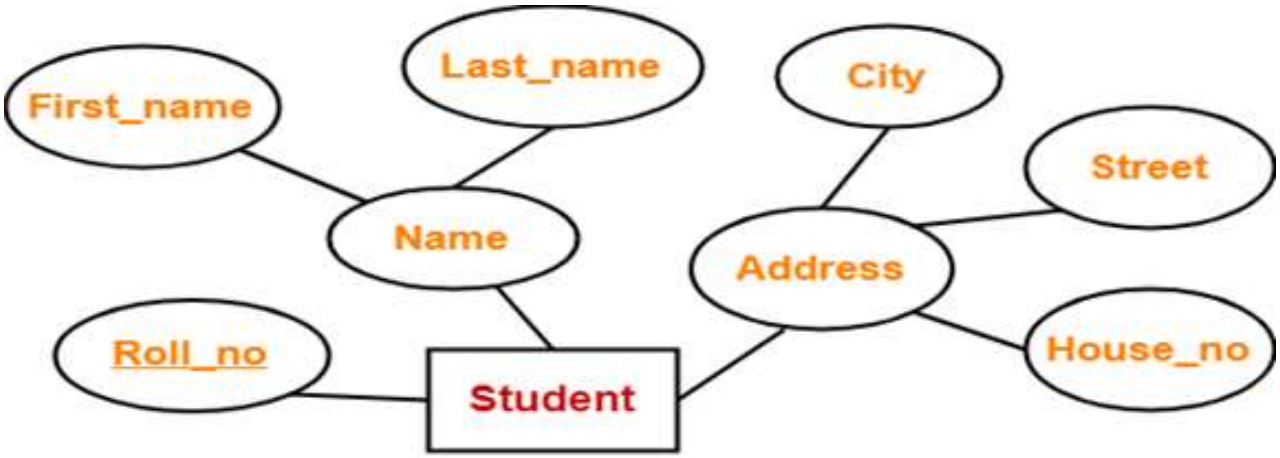
Key attribute ⇒ Primary Key

For Strong Entity Set With Composite Attributes

- A strong entity set with any number of composite attributes will require only one table in relational model

- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.





- What are the Composite Attributes in this example?
  Name and Address
- What are the simple attributes in 'Name' ?
  First_Name, Last_Name
- What are the simple attributes in 'Address' ?
  House_no, Street, City

By considering only simple attributes of the composite attributes in schema design

Roll_no	First_name	Last_name	House_no	Street	City

Schema : Student (Roll_no , First_name , Last_name , House_no , Street , City)

For Strong Entity Set With Multi-Valued Attributes

- A strong entity set with multi-valued attribute will require two tables in relational model
  - One table will contain all the simple attributes with the primary key.
  - Other table will contain the primary key and the multi valued attribute.
  - Create a separate table for each multi-valued attribute.



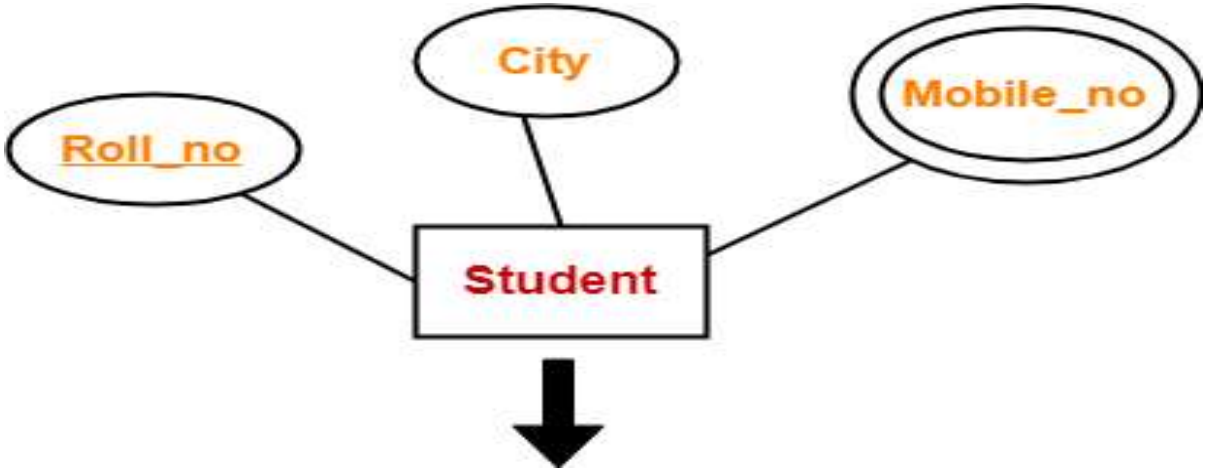
Rollno#	Firstname	Phone_no
1	A	1234
1	A	2345
2	B	1456
2	B	1789



PK – will not allow redundant roll number

separate table for each multi-valued attribute





- Is there a mutivalued attribute?
  Yes – Mobile-no

Roll_no	City

Roll_no	Mobile_no

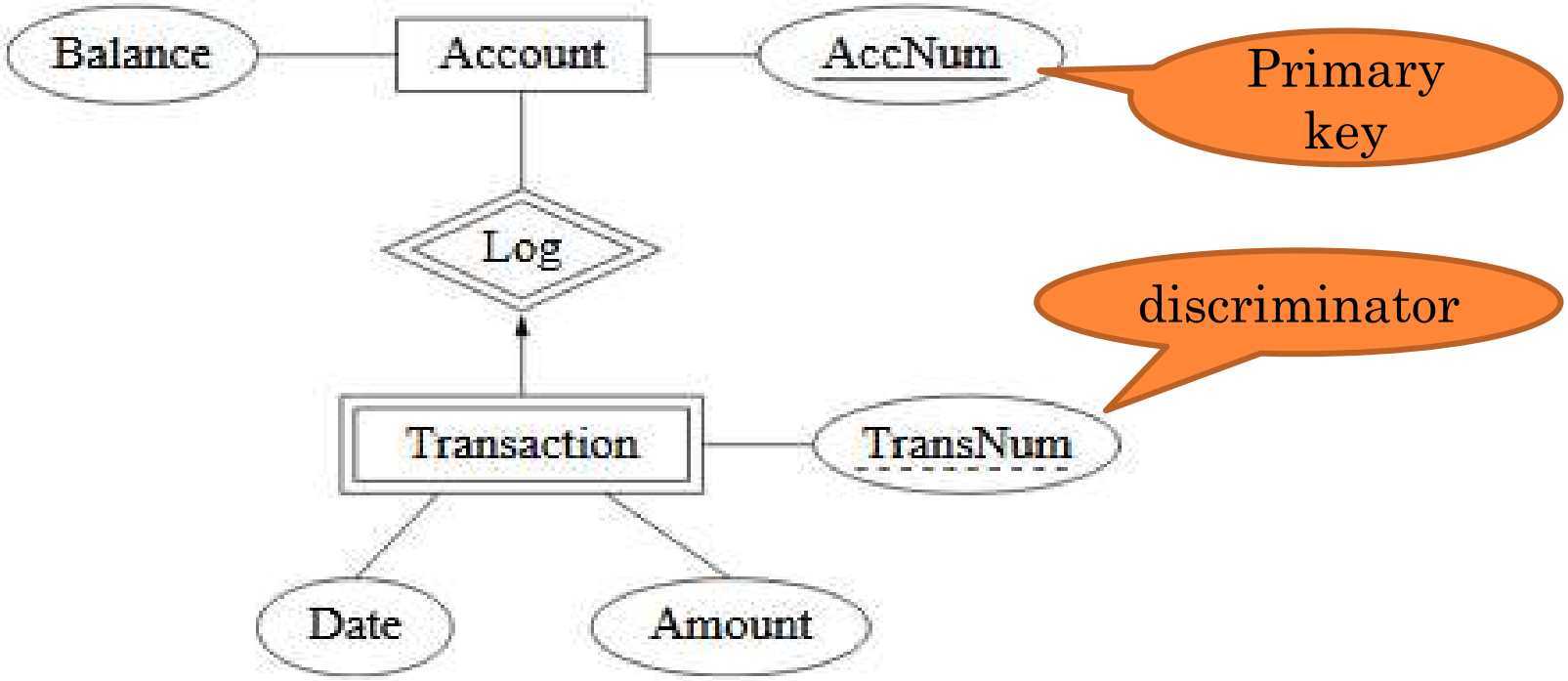
Representation of Weak Entity Sets

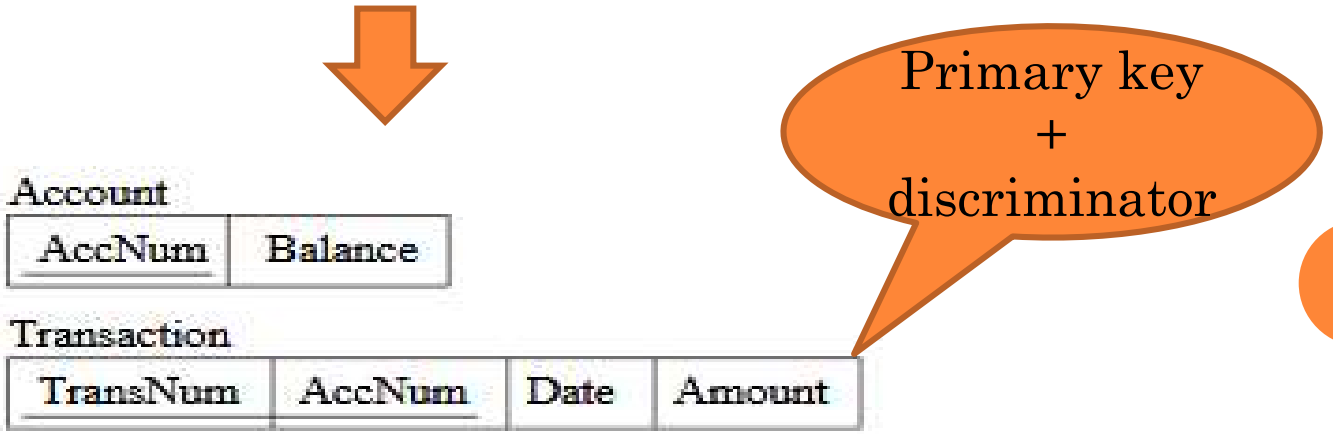
- The entity sets which do not have sufficient attributes to form a primary key are known as **weak entity sets.**
- Weak entity is **depend on strong entity** to ensure the existence of weak entity.
- The discriminator (or) partial key of a weak entity set is the set of attributes that distinguishes among all the entities of a weak entity set.
- **Weak entities** are represented with **double rectangular** box in the ER Diagram and the identifying relationships are represented with double diamond.



CONTD..

A weak entity set becomes a table with all its attributes and also includes a column for the primary key of the identifying strong entity set.





Representation of Relationship Sets

- Let R be a relationship set, let a_1, a_2, …, a_m be the set of attributes formed by the union of the primary keys of each of the entity sets participating in R.

- Let the descriptive attributes (if any) of R be b_1, b_2,…, b_n .

- The representation of relation for the relationship set R can have the attributes as

$$\{a_1, a_2, \ldots, a_m\} \cup \{b_1, b_2, \ldots, b_n\}$$

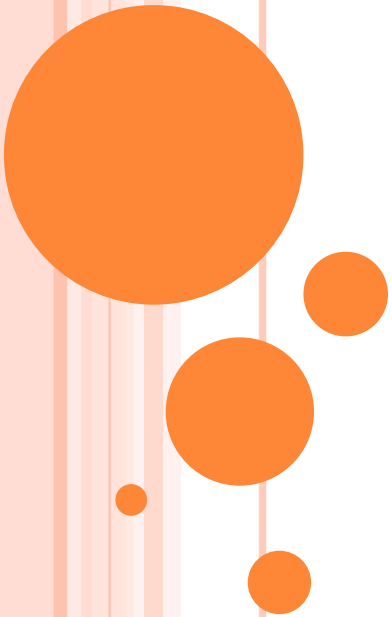


Representation of Relationship Sets (Contd..)

- Selection of Primary key for the relation of relationship set
  - Binary many-to-many relationship
    - The union of the PK attributes from the (all) participating entity sets becomes the PK
  - Binary one-to-one relationship
    - The Primary key of either entity set can be chosen as the PK. The choice can be made arbitrarily
  - Binary many-to-one or one-to-many relationship
    - The Primary key of the entity set on the many side of the relationship set serves as the PK
    - One – serves as FK



SESSION -2



CARDINALITY

Case-01: Binary relationship with cardinality ratio m:n

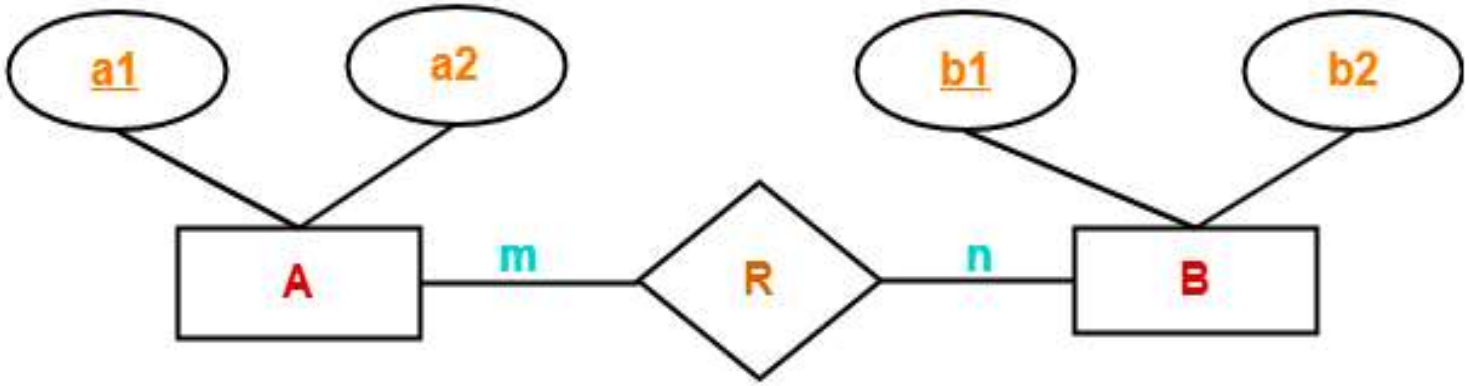
Case-02: Binary relationship with cardinality ratio 1:n

Case-03: Binary relationship with cardinality ratio m:1

Case-04: Binary relationship with cardinality ratio 1:1



Case-01: For Binary Relationship With Cardinality Ratio m:n



Here, three tables will be required-

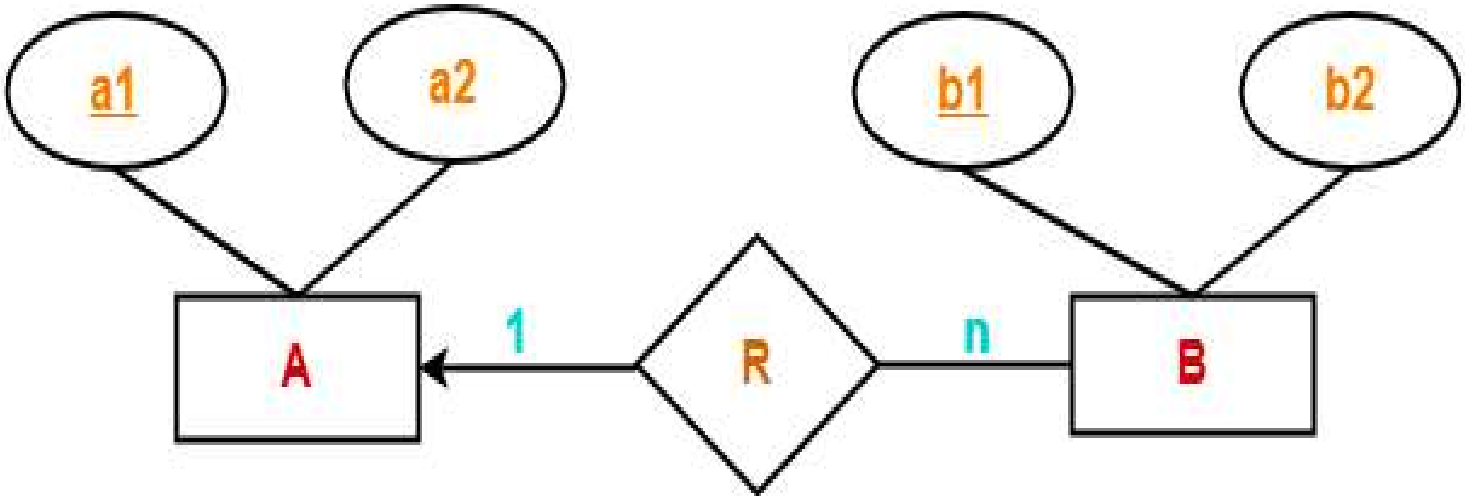
1. A (a1 , a2)
2. R (a1 , b1)
3. B (b1 , b2)



Employee (EmpNo, ...)
Department (Dept_id, ...)

DeputedTo (EmpNo, Dept_id, RecordNo, ...)

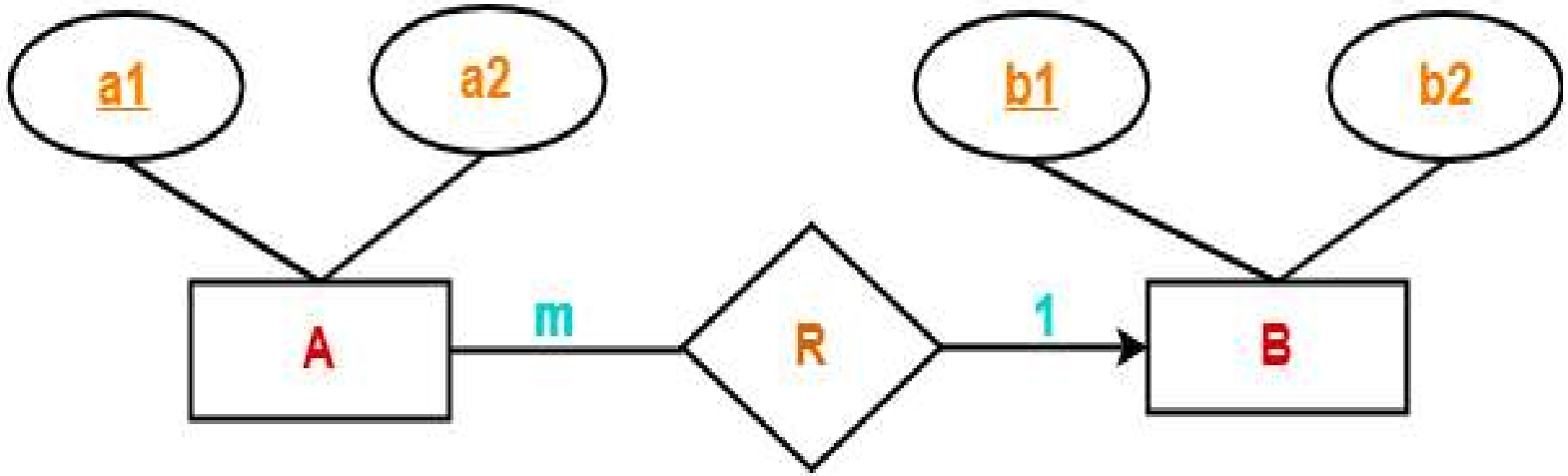
Case-02: For Binary Relationship With Cardinality Ratio 1:n



Here, two tables will be required-

1. A (a1 , a2)
2. BR (a1 , b1 , b2)

Case-03: For Binary Relationship With Cardinality Ratio m:1



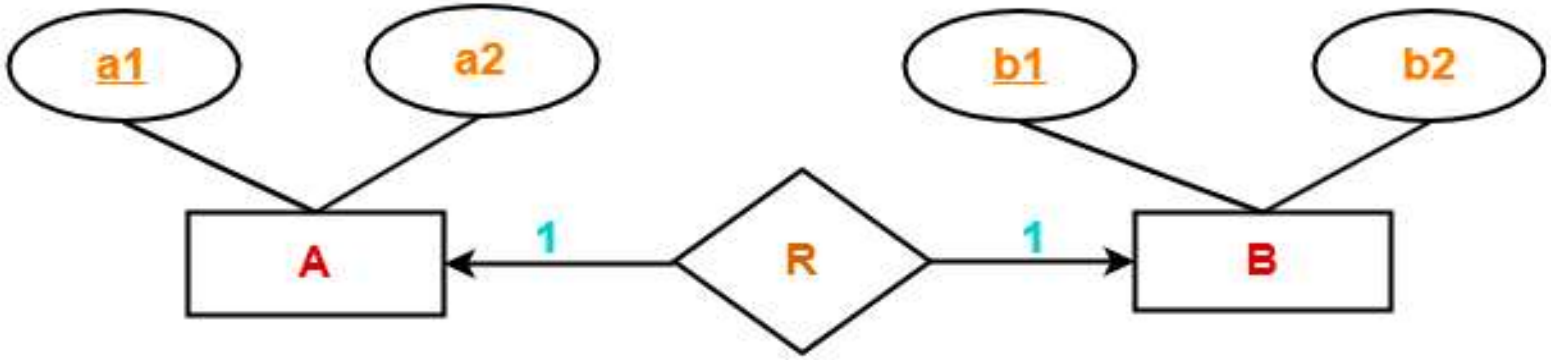
Here, two tables will be required-

1. AR (a1 , a2 , b1)
2. B (b1 , b2)



Employee (EmpNo, Dept_id, …)

Case-04: For Binary Relationship With Cardinality Ratio 1:1



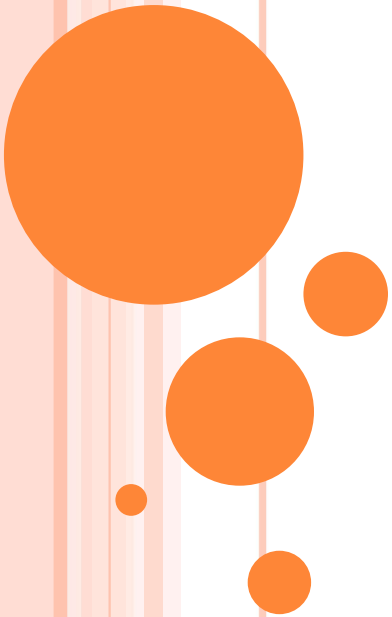
Here, two tables will be required. Either combine 'R' with 'A' or 'B'

Way-01:

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

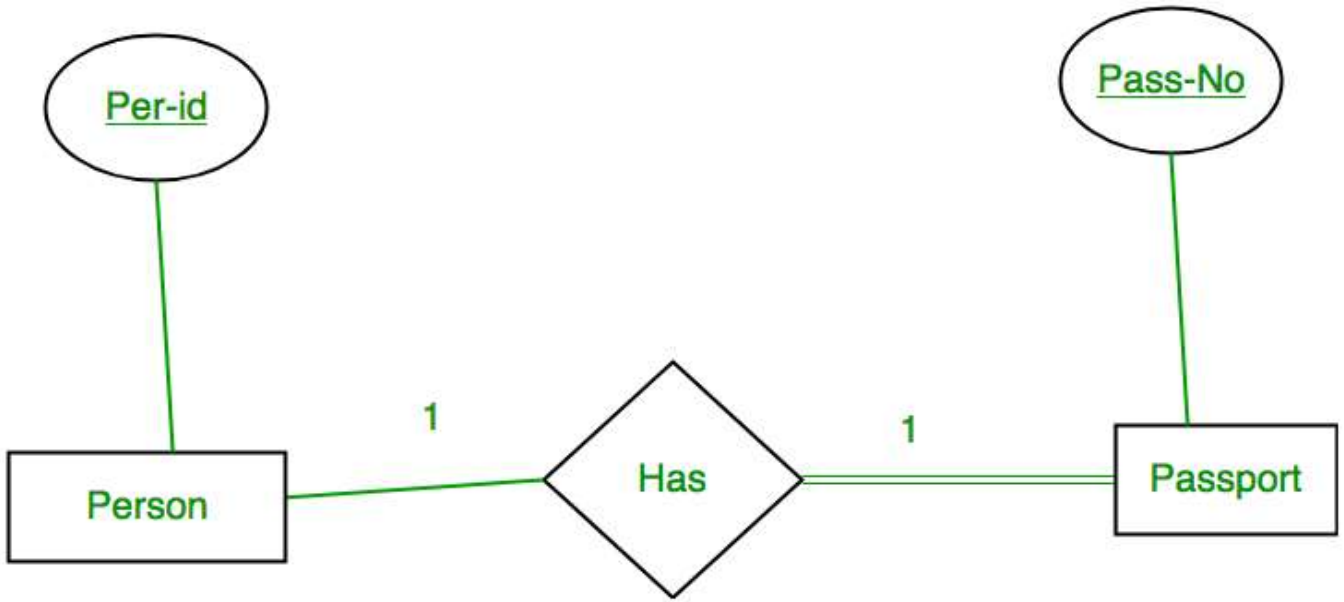
Way-02:

1. A (a1 , a2)
2. BR (a1 , b1 , b2)



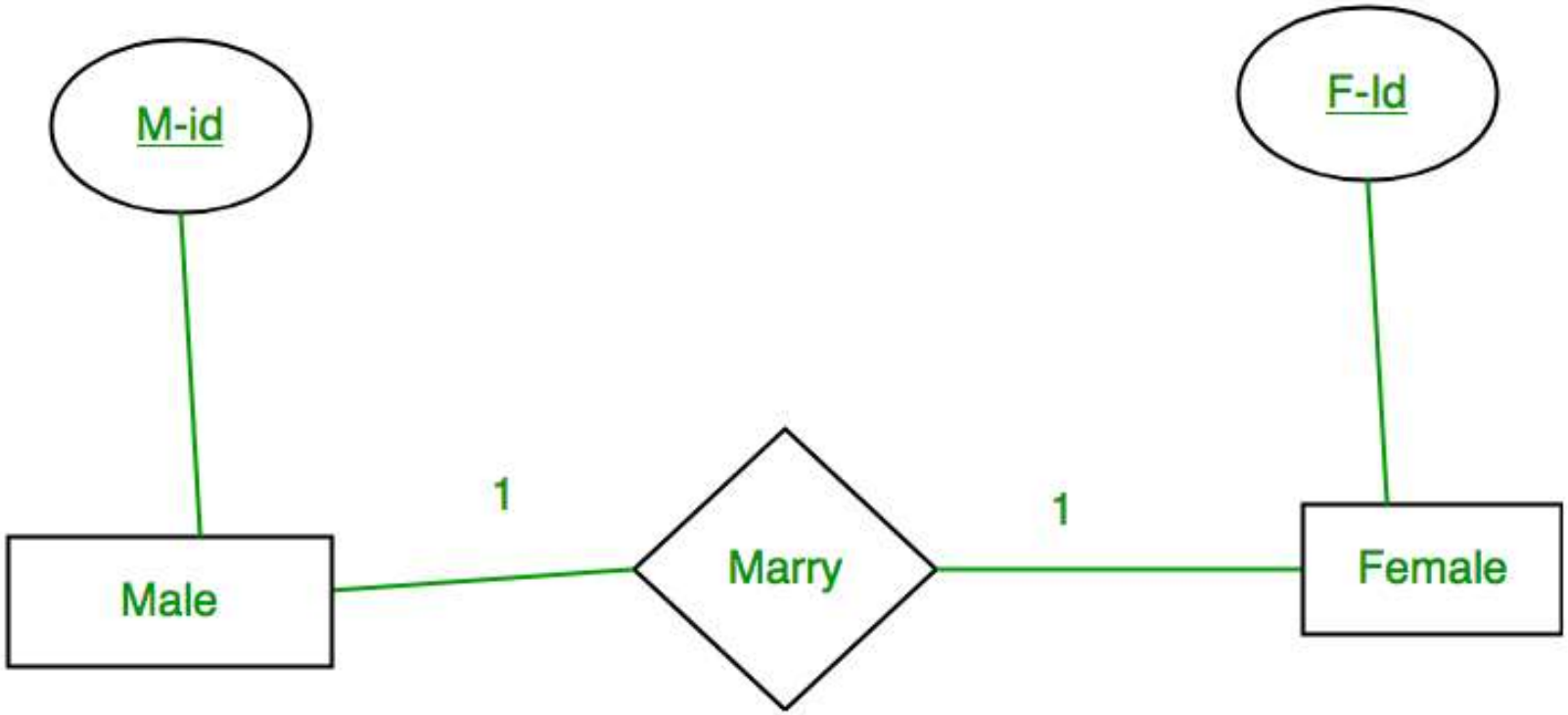
Cardinality with participation

Case 1: Binary Relationship with 1:1 cardinality with TOTAL PARTICIPATION of an entity



Passoport details(per-id, pass no)

Case 2: Binary Relationship with 1:1 cardinality and partial participation of both entities

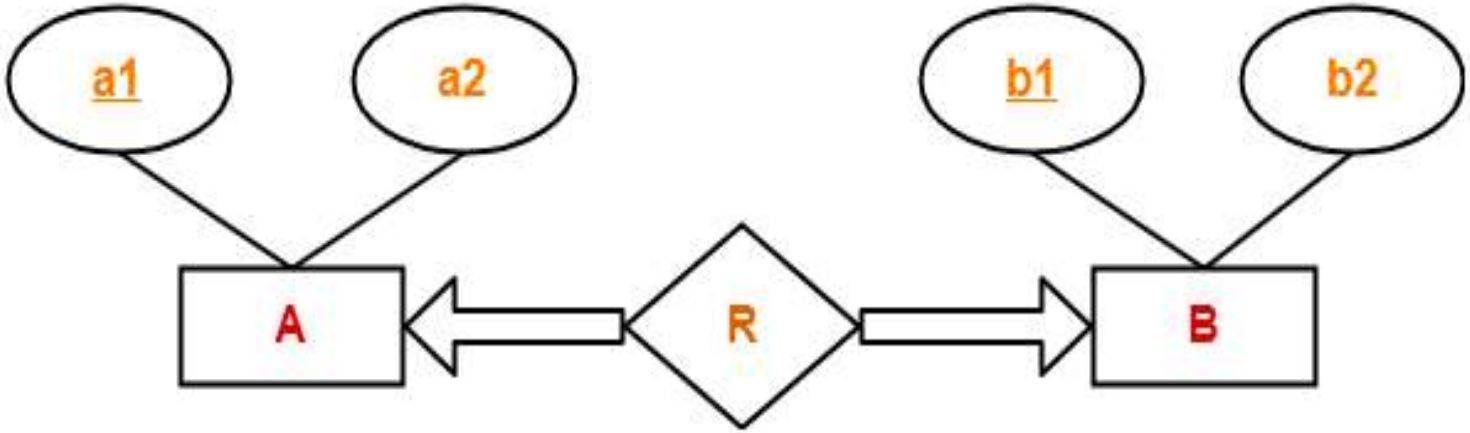


M-Id F-Id

F-Id Other FemaleAttribute

Case-02: For Binary Relationship With Cardinality Constraint and Total Participation Constraint From Both Sides-

If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using only single table.



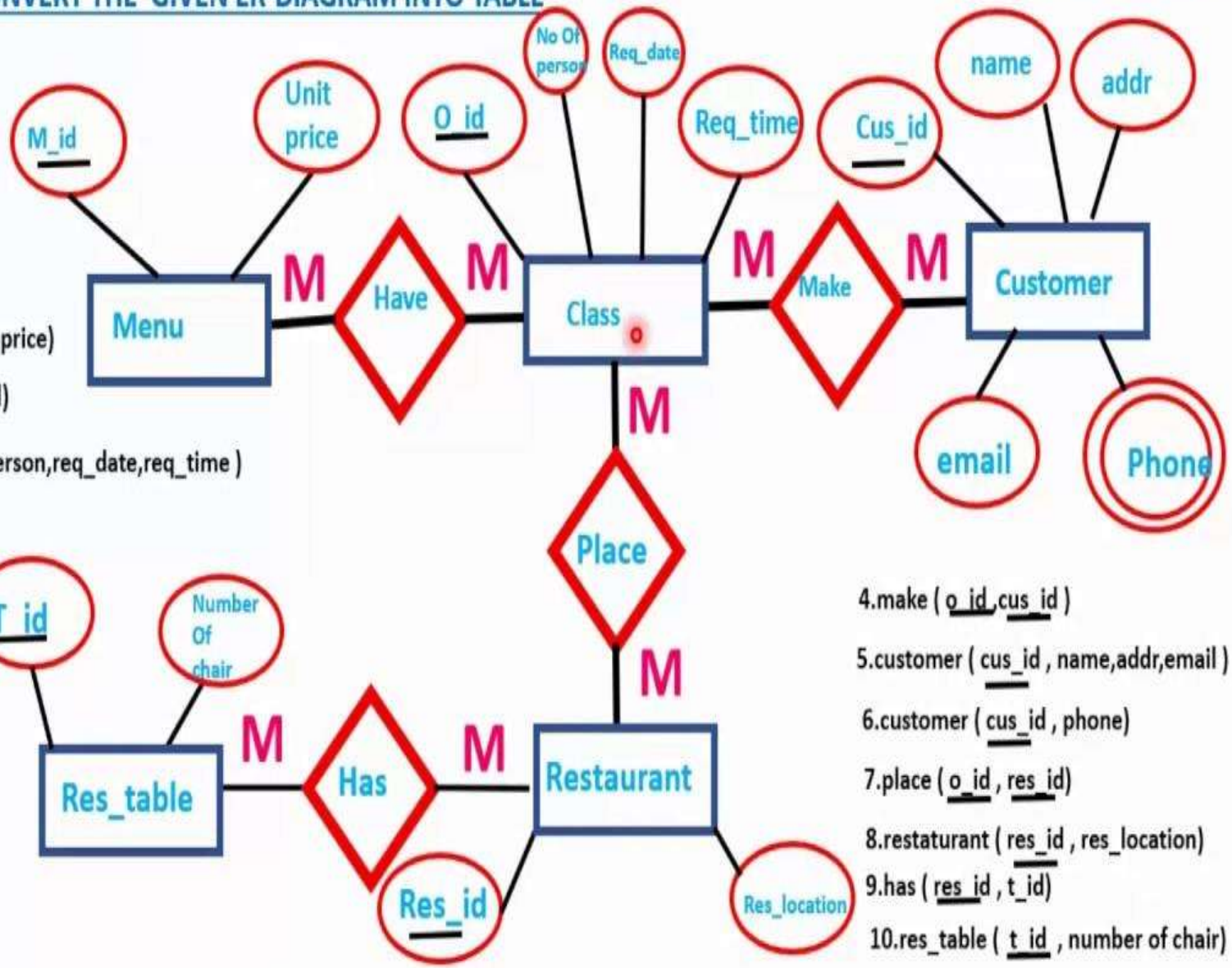
Here, Only one table is required.

- ARB (a1 , a2 , b1 , b2)

CONVERTING ER-DIAGRAM TO TABLE

SUBSCRIBE

EXAMPLE :-2 CONVERT THE GIVEN ER-DIAGRAM INTO TABLE



1.menu (m_id , unit price)

2.have(m_id , o_id)

3.class (o_id,no of person,req_date,req_time)

4.make (o_id,cus_id)

5.customer (cus_id , name,addr,email)

6.customer (cus_id , phone)

7.place (o_id , res_id)

8.restaturant (res_id , res_location)

9.has (res_id , t_id)

10.res_table (t_id , number of chair)

Imagine a **university database** with the following components:

Entities:

1.Student

- Attributes: StudentID (Primary Key), Name, Age

2.Course

- Attributes: CourseID (Primary Key), Title, Credits

3.Instructor

- Attributes: InstructorID (Primary Key), Name, Department

Relationships:

1.Enrolls

- Between: Student and Course
- Type: Many-to-Many
- Attribute: Grade

2.Teaches

- Between: Instructor and Course
- Type: One-to-Many (One instructor can teach many courses)

